

CodeLit

A Standalone LAN-Based Windows Desktop Application for
Secure Exam Lockdown and Automated Programming Assessment

in Philippine Computing Education

Aratia, Brian Joseph B.

Avelino, Robert Emmanuel C.

Perez, Mel Stephen A.

BS Information Technology · University of San Jose-Recoletos

Capstone Proposal

Statement of the Problem

Programming examinations require two critical capabilities that no single platform in Philippine computing education currently provides together.

1



Exam Security

Prevent students from accessing unauthorized resources during live programming assessments through system-wide desktop enforcement.

Security-focused solutions cannot evaluate code

2



Automated Assessment

Evaluate programming solutions efficiently and accurately using hidden test cases, automated grading, and real-time feedback.

Grading systems cannot enforce desktop-level restrictions

Challenges in the Philippine Context

Cloud-dependent examination platforms carry significant risks for Philippine schools.

1



Unreliable Connectivity

Internet reliability remains a major concern across many Philippine universities.

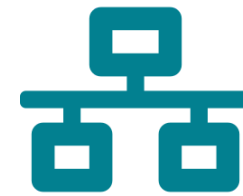
2



Network Interruptions

A single outage can interrupt an entire live programming examination session.

3



Limited Broadband Access

Provincial schools often lack stable high-bandwidth connections needed for cloud tools.

4



Subscription Costs

Budget constraints make recurring software fees unsustainable for many institutions.

5



High-Stakes Risk

Any disruption to an internet-dependent exam can compromise academic fairness.

Existing Solutions

CodeChum

Leading programming assessment platform used in Philippine schools.

- Automated code checking with hidden test cases
- Multi-language: C, C++, Java, Python, JavaScript, C#
- Real-time student monitoring and analytics
- Exam mode with randomized activities

Primary Focus: Automated Programming Assessment

Safe Exam Browser (SEB)

Open-source examination security platform used worldwide.

- Full-screen environment with browser lockdown
- Restricts access to unauthorized websites
- Supports Moodle, Canvas, and Blackboard
- Free, open-source, institution-controlled

Primary Focus: Secure Exam Delivery

Key Gaps in Current Solutions

- ✗ No full desktop-level lockdown in either solution
- ✗ Neither operates offline or on LAN alone
- ✗ No single platform grades code AND secures the OS

- ✗ Cloud dependency creates single points of failure
- ✗ Local data ownership is absent or incomplete
- ✗ Monitoring and grading remain fragmented workflows

Introducing

CodeLit

A single Windows desktop application that combines system-wide exam lockdown with automated programming assessment, operating fully on a local network.

Built for Philippine computing education:

■ System-Wide Lockdown

■ LAN-Only Operation

■ School-Owned Data

■ Automated Code Grading

■ No Internet Required

■ No Subscription Fees

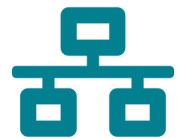


Security + Assessment + Independence

Why CodeLit?

Five core advantages that set CodeLit apart from any existing tool.

01



Fully LAN-Based

Operates entirely on the local network with zero internet dependency. No cloud service outage can disrupt an examination.

02



School-Owned Data

All exam records, student results, and incident reports are stored in a local SQLite database fully under institutional control.

03



No Subscription Fees

A single deployment covers the institution with no recurring costs, making CodeLit accessible to budget-limited schools.

04



Local Code Execution

Uses locally installed GCC and JDK to compile and run student code. No third-party grading service required.

05



True OS-Level Security

Secures the entire Windows environment, not just a browser tab, using Windows API hooks to suppress Alt+Tab, Task Manager, and more.

Key Features

Part 1 of 2

01



System-Wide Exam Lockdown

Uses Windows API hooks to suppress Alt+Tab, Windows Key, Task Manager, copy-paste, and screenshot shortcuts. Secures the entire OS environment, not just a browser session.

02



Live Teacher Dashboard

Real-time monitoring of all student screens over LAN. Sends instant violation alerts to the teacher and logs all security incidents throughout the exam session.

03



Local Code Assessment

Automatically grades student submissions using hidden test cases, process isolation, and time limits. Runs entirely on local GCC and JDK with no third-party service.

Key Features

Part 2 of 2

04



Multi-Type Exam Builder

Supports Multiple Choice, True/False, Fill-in-the-Blank, Short Answer, and Live Coding questions in a single examination. No need for separate tools per question type.

05



Monaco Editor (VS Code-Grade)

Students write code in the same editor used in Visual Studio Code. Familiar syntax highlighting, bracket matching, and autocomplete improve the exam experience.

06



Join by Class Code

Students join sessions using a simple class code with no individual account required. Reduces setup time and removes the need for student registration management.

07



Export Reports

Generates CSV results for grade processing and automated PDF incident reports for academic integrity documentation after every examination session.

How CodeLit Fills the Gaps

Existing Gap	CodeLit Solution
Separate lockdown and grading tools	Unified single platform
Browser-only security enforcement	OS-level Windows lockdown
Internet dependency for exams	LAN-only offline operation
Cloud-based data storage	Local SQLite database
Monthly subscription costs	No recurring fees, single deployment
No automated incident reporting	Automated PDF incident reports
Fragmented teacher and student workflow	End-to-end examination system

Expected Benefits



Academic Integrity

Stronger exam security

Reduced cheating opportunities

Automated incident documentation



Accessibility

Works in low-connectivity areas

Suitable for provincial schools

No student account registration



Cost Efficiency

No monthly subscription fees

Single institutional deployment

Open local toolchain (GCC, JDK)



Data Ownership

Full control of exam records

No third-party cloud storage

Local SQLite database on-premise

CodeLit delivers security, accessibility, and efficiency in one platform built for Philippine computing education.

Conclusion

Current solutions address only part of the problem. CodeLit bridges the gap.

1

System-Wide Lockdown

Secures the entire Windows environment

2

Automated Code Assessment

Hidden test cases, auto-grading, time limits

3

LAN-Only Deployment

Zero internet dependency for exam delivery

4

Offline Operation

Immune to network outages and cloud failures

5

School-Owned Data

Records stay within the institution always



CodeLit

Secure · Automated
Independent

Thank You

Questions and Discussion

Aratia, Brian Joseph B.
Avelino, Robert Emmanuel C.
Perez, Mel Stephen A.

*BS Information Technology
University of San Jose-Recoletos*

CodeLit